MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RADIAL BEAM SWITCHING

TYPE 6700

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Permanent magnet, high vacuum, 10 outputs (see note 1)

See figure 1

Mounting position: Any

Weight: 7.5-ounces nominal

ABSOLUTE RATINGS:

Parameter:	Ef	Ehk	Eta	Esp	Ecc	Rta	Rsp
Unit:	V	V	Vdc	Vdc	Vdc	Ohms	Ohms
Maximum:	6.9	±100	300	300	300	---	---
Minimum:	5.7	---	---	---	---	---	---
TEST CONDITIONS:	6.3	---	100	100	+25	4,700 ±1%	100,000 ±1%

ABSOLUTE RATINGS:

Parameter:	Eg-sig	F	TA	P/ta	Ik	Barometric pressure, reduced
Unit:	V	MHz	°C	W	mA	mmHg
Maximum:	---	Note 10	+85	1	20	55
Minimum:	---	---	-65	---	---	---
TEST CONDITIONS:	-50	---	---	---	---	---

GENERAL:

Qualification - Not required

(D) denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:05Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1						
---	Counting voltage (range 1)	5, 6, 13	4 Hz; Eta = Esp = 90 to 110 Vdc	---	---	---	---
---	Counting voltage (range 2)	5, 6, 13	10 kHz; Eta = Esp = 90 to 110 Vdc	---	---	---	---
1256	Electrode current (target)	6, 13	Individual targets 10 kHz	Ita	4.7	6.3	mAdc
	Quality conformance inspection, part 2						
---	Interelectrode leakage	-	Any element to all others (except cathode) 400 Vdc	---	---	5.0	µAdc
1256	Electrode current (spade 1)	4, 12	Esig = 0	Isp1	---	40	µAdc
1256	Electrode current (spade 2)	4, 12	Eta = Esp = 0 to 200 V; Esig = 0	Isp2	---	300	µAdc
---	Magnetic secureness	15		---	---	---	---
1031	Sweep-frequency vibration	1, 2, 6, 14	F = 55 to 500 Hz; tube operating at 10 kHz; t = 300 sec	---	---	---	---
(D) ---	Shock	1, 2, 16	No voltages applied; 450 G at 0.8 ms; half-sine wave	---	---	---	---
---	Counting voltage (range 3)	5, 7, 8, 13	1 MHz; Eta = Esp = 90 to 110 Vdc; Eg-sig = -60 V	---	---	---	---
2126	Envelope strain	-		---	---	---	---
1002	Barometric pressure, reduced	-	55 ±5 mmHg; 300 Vdc between targets and all other elements	---	---	---	---
1301	Heater current	-		If	270	330	mA
1336	Heater-cathode leakage	-	Ehk = ±100 V	Ihk	---	20	µAdc
	Quality conformance inspection, part 3						
---	Life test (1)	-	Group C; dynamic; 120 counts/sec sine-wave excitation; t = 1,000 hours (min)	---	---	---	---
---	Life test (2)	-	Group C; static; beam formed and locked on any one target; t = 1,000 hours (min)	---	---	---	---
---	Life test (3)	-	Group C; cutoff; Eta = Esp = Ecc = 200 V; Rk = 10,000 ohms; t = 1,000 hours (min)	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:05Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 -Continued						
---	Life-tests (1), (2), and (3) end points:						
1256	Electrode current (spade 1)	-		Isp1	---	50	µAdc
1256	Electrode current (spade 2)	-		Isp2	---	400	µAdc
1256	Electrode current (target, individual)	-		Ita	4.7	6.3	mAdc
---	Counting voltage (range 1)	-	Eta = Esp = 90 to 110 Vdc	---	---	---	---
---	Counting voltage (range 2)	-	Eta = Esp = 90 to 110 Vdc	---	---	---	---
---	Electrical noise (individual targets)	3, 6, 11, 12	10 kHz	---	---	500	mVac
---	Light sensitivity	9, 11, 12		ΔIk	---	0.1	mAdc
1027	Temperature cycling (nonoperating)	11	5 cycles	---	---	---	---

NOTES:

1. In order to avoid possible operational changes in the characteristics of this tube, do not strike the magnet, or place the tube in close proximity (less than 2 inches) to magnetic materials and fields, without proper shielding.
2. Magnet only shall be held rigidly. Socket shall float in order not to disturb the magnetic field alinement of the tube.
3. Noise is the rms noise voltage across the individual target resistors.
4. Tube shall be in cutoff condition.
5. Tube shall count 10 at specified frequency and throughout counting voltage range of Eta and Esp.
6. (D) Pulse width (measured at 100 percent point) of 90 µs minimum applied alternately to odd and even grids. Grid swing from +25 V to -25 V with respect to cathode.
7. At 1 MHz, individual spade stem leads capacity shall not exceed 10 pF with tube in socket.
8. At 1 MHz, pulse width (measured at 90 percent point) of 0.67 µs minimum applied alternately to odd and even grids. Grid swing from +25 V to -35 V with respect to cathode.
9. Lock beam in position "0". Check cathode current. Bring photo-flood lamp (RFL-2, or equivalent; 500 watts) within 6 inches of tube in any direction. Recheck cathode current. Variation in current shall not exceed maximum specified.
10. Tube is tested up to 1 MHz but may operate to 5 MHz under optimum circuit conditions.
11. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.
12. Test in circuit shown on figure 2.
13. Test in circuit shown on figure 3.
14. At the completion of the vibration test, the tube shall count 10 at the specified test conditions at 10 kHz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:05Z
Check the source to verify that this is the current version before use.

NOTES: -Continued

15. Thrust between glass envelope and magnet shall be 30-pounds minimum. The tube shall be immersed in water at 50°C for 42 hours, and then removed and cooled for 1 hour at room temperature before testing.

16. At the completion of the shock test, the tube shall count 10 at the specified test conditions at 10 kHz.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

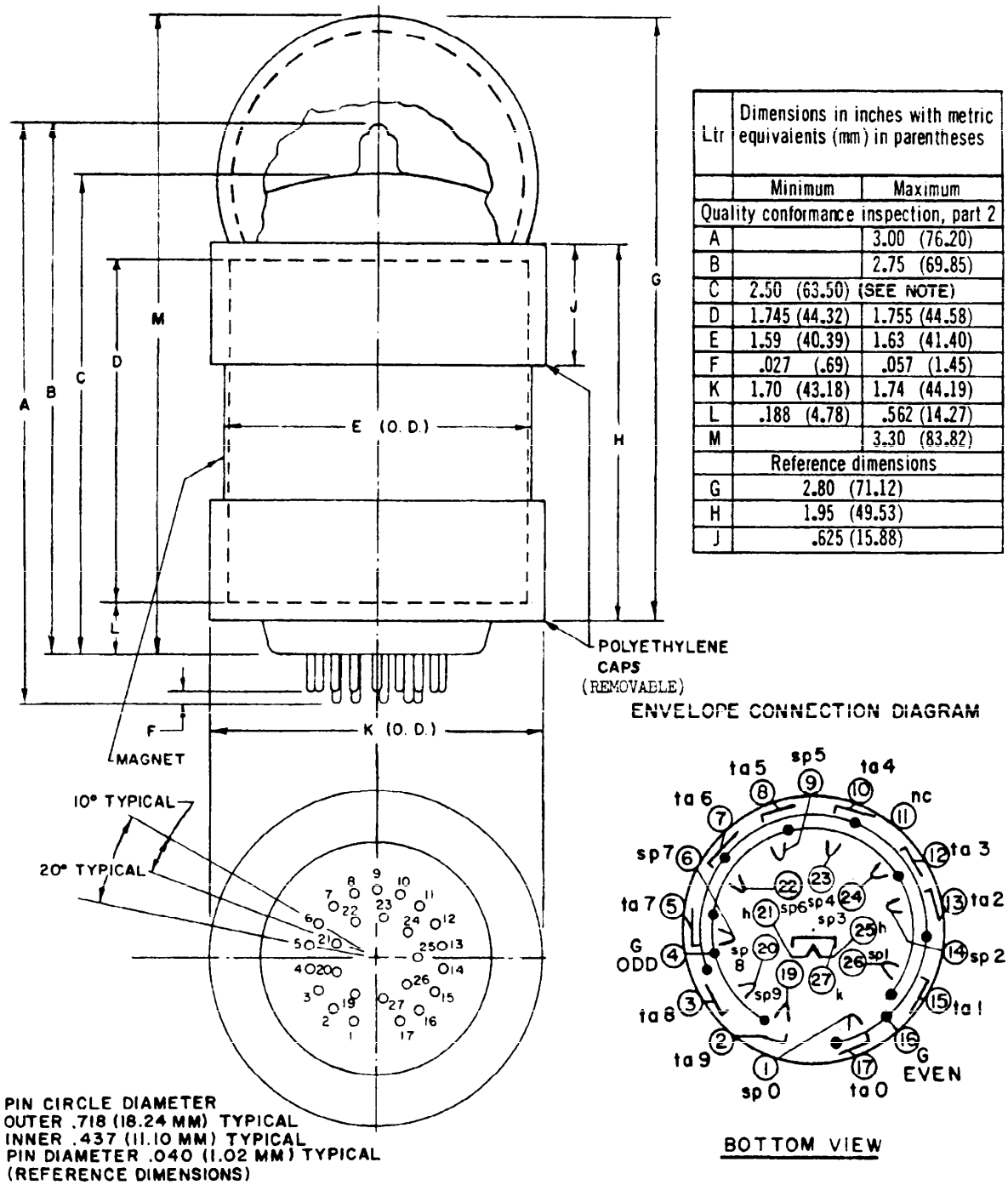
User activities:
Army - MU
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3055)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:05Z
Check the source to verify that this is the current version before use.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		3.00 (76.20)
B		2.75 (69.85)
C	2.50 (63.50) (SEE NOTE)	
D	1.745 (44.32)	1.755 (44.58)
E	1.59 (40.39)	1.63 (41.40)
F	.027 (.69)	.057 (1.45)
K	1.70 (43.18)	1.74 (44.19)
L	.188 (4.78)	.562 (14.27)
M		3.30 (83.82)
	Reference dimensions	
G	2.80 (71.12)	
H	1.95 (49.53)	
J	.625 (15.88)	

NOTE: As determined by ring gage of .437 (11.10 mm) I.D.

FIGURE 1. Outline drawing of electron tube type 6700.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:05Z
Check the source to verify that this is the current version before use.

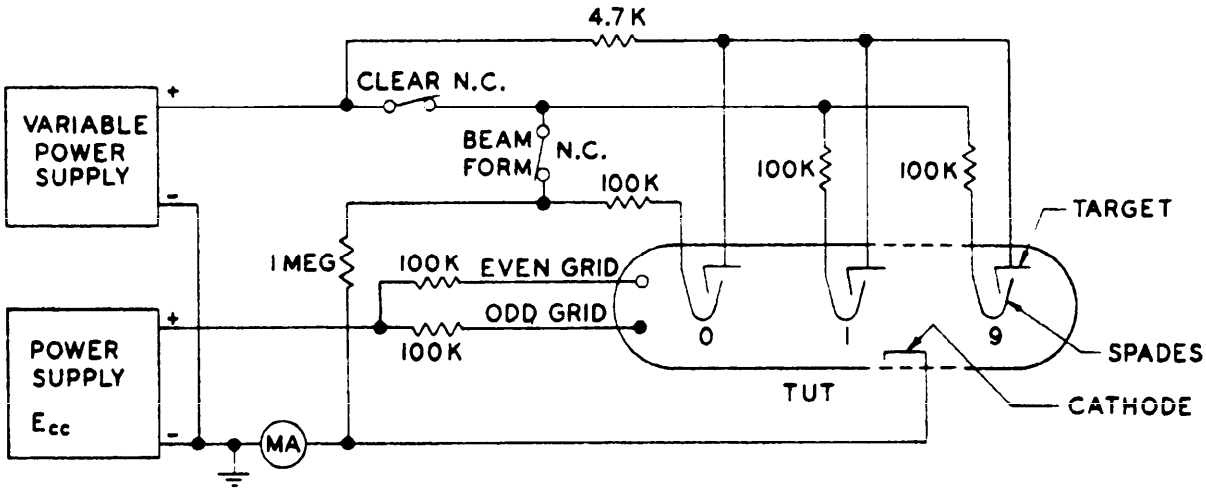


FIGURE 2. Test circuit.

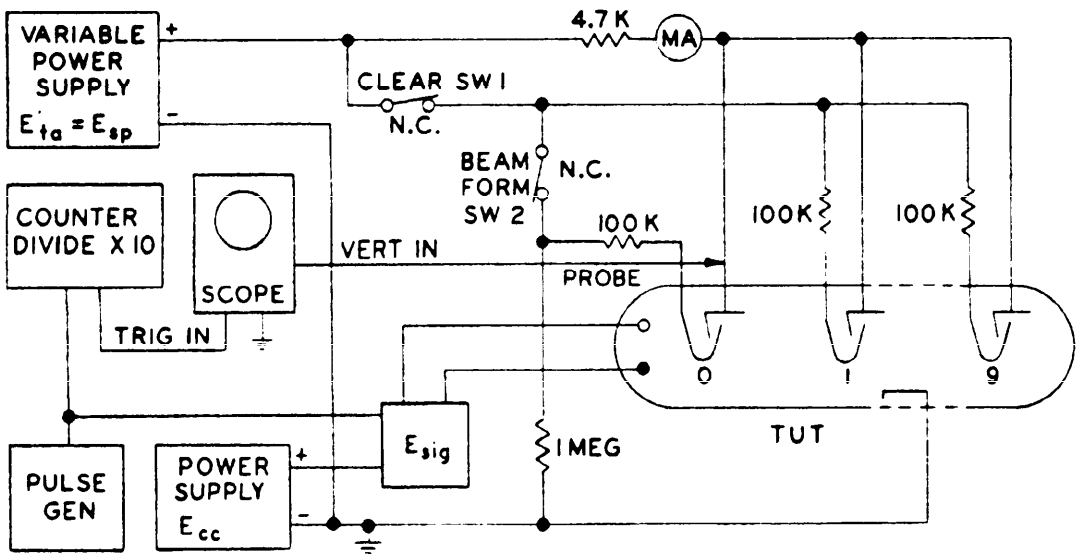


FIGURE 3. Test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:05Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:05Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:05Z
Check the source to verify that this is the current version before use.